

Shopping Cart REST API Documentation

Cart Endpoints

Retrieve Active Cart

Endpoint: `GET /carts/active`

Response Body:

```
{
  "cartId": "uuid-string",
  "userId": "uuid-string",
  "items": [
    {
      "itemId": "uuid-string",
      "productId": "uuid-string",
      "name": "Product Name",
      "quantity": 2,
      "pricePerUnit": 49.99,
      "subtotal": 99.98
    }
  ],
  "totalAmount": 99.98,
  "status": "ACTIVE"
}
```

Add Item to Cart

Endpoint: `POST /carts/items`

Request Body:

```
{  
  
  "productId": "product-uuid",  
  
  "quantity": 1  
  
}
```

Update Item Quantity

Adjusts the quantity of a specific item within the cart. This is typically used on the cart summary page to increment or decrement counts.

Endpoint: `PATCH /carts/items/{itemId}`

Request Body:

```
{  
  
  "quantity": 3  
  
}
```

Remove Item from Cart

Deletes a specific line item from the cart entirely.

Endpoint: `DELETE /carts/items/{itemId}`

Success Response: `204 No Content`

Clear Cart

Removes all items from the active cart. This action is usually performed when a user chooses to empty their cart or after a successful checkout session.

Endpoint: `DELETE /carts/active/clear`

2. Database Schema

Combining the detailed constraints from your digital ER diagrams (specifically the second one, which represents a highly scalable structure) with the notes from your notebook, here is the normalized schema:

Core Tables

Table Name	Column	Data Type	Constraints / Relations
USER	id	INT	Primary Key
	name	VARCHAR	
	email	VARCHAR	Unique
	password	VARCHAR	Hashed
	created_at	TIMESTAMP	Default CURRENT_TIMESTAMP
CATEGORY	id	INT	Primary Key
	name	VARCHAR	
	description	TEXT	
PRODUCT	id	INT	Primary Key
	category_id	INT	Foreign Key - CATEGORY(id)
	name	VARCHAR	
	price	DECIMAL	
	stock	INT	

Table Name	Column	Data Type	Constraints / Relations
	is_active	BOOLEAN	
CART	id	INT	Primary Key
	user_id	INT	Foreign Key -USER(id)
	status	VARCHAR	
	coupon	VARCHAR	Nullable (from your first diagram)
CART_ITEM	id	INT	Primary Key
	cart_id	INT	Foreign Key - CART(id)
	product_id	INT	Foreign Key -PRODUCT(id)
	quantity	INT	Default 1
	price_at_addition	DECIMAL	To lock in price if product price changes

